

DSA210 Term Project Proposal

Weather Conditions and Goal Scoring in Beşiktaş Matches

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This project investigates whether weather conditions have a measurable effect on the number of goals scored in Beşiktaş matches. Football is played under highly variable environmental conditions, and factors such as rain, temperature, and wind may influence match tempo, player comfort, and finishing quality. The main question of the project is whether adverse or changing weather conditions are associated with differences in total goals scored in Beşiktaş games.

The project will combine two publicly available datasets. Match data will be collected from football statistics sources such as FBref and Football-Data.co.uk, including the match date, opponent, home/away status, final score, and total goals in the match. Historical weather data will be collected through the Open-Meteo API and matched to each game by date, approximate kickoff time, and match location. The enriched dataset is expected to cover approximately the last 5 to 10 seasons of Beşiktaş league matches. Since each season contains roughly 34 to 40 matches, the final dataset is expected to include about 170 to 400 observations, depending on data completeness and consistency.

The main hypothesis is: rainy weather significantly affects the total number of goals scored in Beşiktaş matches. To study this, I will first perform exploratory data analysis using descriptive statistics, visualizations, and comparisons between rainy and non-rainy matches. Then I will apply hypothesis testing, such as an independent samples t-test or a nonparametric alternative if needed, to evaluate whether goal counts differ significantly across weather conditions. If the data size and quality are sufficient, I may also apply simple regression or machine learning models to estimate the contribution of weather variables such as precipitation, temperature, and wind speed to goal-related outcomes.